FasterCap 3D Input Files

Conductor definitions ('C' statement)

```
Syntax: C <file> <outperm> <xoffset> <yoffset> <zoffset> [+]
```

The 'C' element at the beginning of a line defines a conductor. The geometry of the conductor surface is further specified in the file `<file>`. The conductor is assumed embedded in an homogeneous dielectric medium with relative permettivity `<outperm>`. The relative permittivity `<outperm>` can be complex valued, in the format *ere*-j*eim*, where *ere* is the real part of the complex permittivity value, and *eim* is the imaginary part. The conductor can be translated with respect to the coordinates defined in `<file>` by an offset *(xoffset, yoffset, zoffset)*, thus allowing to reuse the same geometric definitions multiple times. The optional `+` argument is used to merge the current conductor with the next one, thus treating them a single conductor. With this option, panels with the same conductor name in different files will belong to the same conductor (see panels definitions for more information about conducor naming).

Example 1:

```
C cube.txt    1.000000     2.0 0.0 0.0
```

This input file fragments specifies a conductor whose geometry is defined in the file `cube.txt`, embedded in a dielectric with permettivity equal to one (e.g. air), and translated to the position *(2.0, 0.0, 0.0)*.

Example 2:

```
C cube.txt    1.000000     0.0 0.0 0.0 +
C cube.txt    1.000000     2.0 0.0 0.0
```

This input file fragments specifies a conductor made of two cubes in free air. It is considered a single conductor since the first statement ends by `+`, thus collating the two elements; i.e. the two cubes are considered short-circuited together.

Example 3:

```
C cube.txt    3.0-j0.02     2.0 0.0 0.0
```

This input file fragments specifies a conductor whose geometry is defined in the file `cube.txt`, embedded in a dielectric with complex permettivity equal to 3.0-j0.02 (e.g. with real part equal to 3.0 and imaginary part equal to 0.02), and translated to the position *(2.0, 0.0, 0.0)*.